

TERRALINK

USER MANUAL

LoRa Mesh-Based Emergency Communication System

1. About TerraLink

TerraLink is an emergency communication system designed for remote and disaster-prone areas where conventional communication may be unavailable. It uses **LoRa wireless communication, GPS, ESP32 and relay nodes** to carry an emergency SOS alert from the user device to a receiver/rescue station.

2. What the User Can Do

- **Trigger an SOS** using the touch sensor when emergency assistance is required.
- **View system status** on the OLED display.
- **Transmit emergency information** through the LoRa network.
- **Send location information** obtained from GPS when available.
- **Use relay-based communication** so the alert can travel through multiple nodes.

3. Main Parts of the System

Part	Purpose
ESP32 Controller	Controls the TerraLink device and manages emergency operation.
TTP223 Touch Sensor	Activates the SOS request.
SX1278 LoRa Module	Provides wireless communication between TerraLink nodes.
GPS Module	Provides location coordinates for the emergency packet.
SH1106 OLED	Displays wake-up, SOS and transmission status.
Buzzer	Provides an audible indication during operation.
Relay Nodes	Forward the emergency packet toward the receiver/rescue station.
Battery / Power Unit	Provides portable power to the system.

4. Before Using TerraLink

- Make sure the battery/power connection is secure.
- Ensure the LoRa antenna is connected before transmission.
- Switch ON the required sender, relay and receiver nodes.
- Keep the GPS unit in an open area when location acquisition is required.
- Check that the OLED and touch sensor are connected and functioning.

5. Starting the System

Step 1: Switch ON the TerraLink sender and required network nodes.

Step 2: Allow the ESP32 and connected modules to initialize.

Step 3: Observe the OLED for the system status.

Step 4: Keep the sender ready for an emergency SOS trigger.

Step 5: When no emergency is active, the system may use its configured low-power/sleep operation.

6. Communication Flow

User/Sender → Relay Node 1 → Relay Node 2 → Receiver / Rescue Station

When an SOS is triggered, TerraLink prepares an emergency packet. The packet can contain node identification, SOS status, GPS location and other system information. Relay nodes forward the packet until it reaches the receiver.

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USER OPERATION • STATUS • TROUBLESHOOTING

7. How to Send an SOS Alert

Step 1 — Activate: Touch the **TTP223 SOS sensor** to initiate the emergency request.

Step 2 — Wake: If the sender is in its low-power state, the ESP32 wakes and begins emergency processing.

Step 3 — Location: The GPS module provides location information when a valid position is available.

Step 4 — Prepare: TerraLink creates the emergency packet with the required information.

Step 5 — Transmit: The LoRa module sends the emergency packet.

Step 6 — Forward: Relay Node 1 and Relay Node 2 forward the packet toward the receiver.

Step 7 — Confirm: Observe the OLED/status indication for the transmission stage.

8. Understanding OLED / Status Messages

Display / Indication	Meaning
TERRALINK AWAKE	The sender has awakened and started emergency operation.
SOS / Emergency	An SOS event has been detected and is being processed.
ALERT TRANSMITTED	The emergency alert has entered the LoRa transmission process.
Deep Sleep / Low Power	The sender is in its configured low-power state when no SOS is active.

9. Troubleshooting

Problem	What to Check
OLED is blank	Check power, I2C wiring, connections and display initialization.
Touch does not trigger SOS	Check TTP223 power/signal connection and touch operation.
GPS location unavailable	Move to an open area and allow time for GPS acquisition.
Alert is not received	Check sender, relay and receiver power, antenna and LoRa connections.
Relay does not forward	Check relay power, LoRa configuration and forwarding logic.
Sender does not wake	Check the touch input and ESP32 wake-up configuration.

10. Safety & Maintenance

- Do not reverse or short-circuit the battery connections.
- Use the specified power arrangement for TerraLink hardware.
- Keep the LoRa antenna connected while transmitting.
- Protect the electronics from water, excessive heat and physical damage.
- Inspect wiring, battery, antenna, relay nodes and receiver connections before demonstrations.
- TerraLink is a communication aid and does not replace official emergency services.

11. Quick User Checklist

Before SOS	During SOS
■ Power ON sender and required nodes ■ Check antenna ■ Check OLED ■ Keep GPS in suitable view	■ Touch SOS sensor ■ Observe TERRALINK AWAKE ■ Wait for transmission ■ Confirm ALERT TRANSMITTED/status

TerraLink — Simple operation. Multi-hop communication. Emergency alert delivery.

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